

Doing the $+$ prescription first

Can the p^+ integral be done with no transverse integration at all? — yes, and here is why

$$S \equiv p^+ + k^+, \quad r \equiv y' - y, \quad \kappa \equiv \frac{k \cdot r}{k^+}, \quad P_V \equiv V - k^+$$

Yes. In this row p^+ touches the transverse coordinates in exactly one place — the phase $e^{-i\kappa p^+}$, which involves y and y' and nothing else. Every kernel, the whole z integral, and the entire colour structure are p^+ -independent, so they are a **spectator factor** that pulls straight out of the p^+ integral. The $+$ prescription therefore acts on a pure phase. Verified: the two integrations commute to the accuracy of the quadrature (§2), and the $p^+ \rightarrow 0$ subtraction term is still transverse-convergent (§3). **Two things to keep in mind:** the $\frac{2}{k^+}$ term is only zero *under* $\int d^2z$, so you must carry it (§4); and the z integral is still infrared divergent, so ϵ is still required eventually — the $+$ prescription does not remove that (§5).

1. The form you can write down

Keeping every transverse coordinate un-integrated:

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\mathcal{C}\mathcal{C}} = \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{1}{k^+} \int_0^{\vee-k^+} \frac{dp^+}{2\pi} \int_{x,x',y,y',z} e^{-ik \cdot (y'-y)} e^{-i\kappa p^+} \mathcal{W}^{mm'kk'} \mathcal{B}_{mm'kk'}(p^+) \mathcal{C}$$

with the purely transverse factor, the purely longitudinal bracket, and the colour

$$\begin{aligned} \mathcal{W}^{mm'kk'} &= \frac{(y-z)^m (y'-z)^{m'} (x-y)^k (x'-y')^{k'}}{(y-z)^2 (y'-z)^2 (x-y)^2 (x'-y')^2}, \\ \mathcal{B}_{mm'kk'}(p^+) &= \frac{d_{\perp} p^+}{(k^+ + p^+)^2} \delta_{km} \delta_{k'm'} + \frac{2}{k^+} (\delta_{mk'} \delta_{m'k} - \delta_{m'k'} \delta_{km}) + \frac{p^+}{k^{+2}} \delta_{mm'} \delta_{kk'} + \frac{1}{p^+} \delta_{mm'} \delta_{kk'}, \\ \mathcal{C} &= N_c [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x), \end{aligned}$$

and the single replacement, understood under $\int_0^{\vee-k^+} dp^+$:

$$\frac{1}{p^+} \longrightarrow \left[\frac{1}{p^+} \right]_+ + \delta(p^+) \log \frac{\vee - k^+}{\Lambda}, \quad \int_0^{P_V} dp^+ \left[\frac{1}{p^+} \right]_+ h(p^+) \equiv \int_0^{P_V} \frac{dp^+}{p^+} [h(p^+) - h(0)].$$

Because $\mathcal{W}$ and $\mathcal{C}$ carry no p^+ , the function the prescription acts on is just $h(p^+) = e^{-i\kappa p^+}$, with $h(0) = 1$. Explicitly, for the $\frac{1}{p^+}$ term:

$$\int_{\Lambda}^{\vee-k^+} \frac{dp^+}{p^+} e^{-i\kappa p^+} \mathcal{W}^{mm'kk'} \delta_{mm'} \delta_{kk'} \mathcal{C} = \left\{ \log \frac{\vee - k^+}{\Lambda} + \int_0^{\vee-k^+} \frac{dp^+}{p^+} [e^{-i\kappa p^+} - 1] \right\} \mathcal{W}^{mm'kk'} \delta_{mm'} \delta_{kk'} \mathcal{C}.$$

Checked numerically against the direct integral at fixed transverse points: agreement to 1.2×10^{-9} with $\Lambda = 10^{-9}$, i.e. the identity is exact up to the expected $O(\kappa\Lambda)$ remainder. The other three terms of $\mathcal{B}$ are regular at $p^+ = 0$ and need no prescription — just integrate them from 0.

2. The two integrations commute

Route A: do p^+ first (with the $+$ prescription), then z numerically. Route B: do z first in closed form, then p^+ . At a generic configuration, with the z grid refined:

z grid	route A (p^+ first, then z)	rel. difference from route B
2000×360	$-139.154067 + 311.789432 i$	1.29×10^{-3}
4000×720	$-139.090478 + 311.648494 i$	8.35×10^{-4}
8000×1440	$-139.022240 + 311.493456 i$	3.38×10^{-4}
16000×2880	$-138.983399 + 311.408011 i$	6.28×10^{-5}
route B (exact)	$-138.975156 + 311.388262 i$	—

Monotone convergence onto the exact value: the order of integration does not matter.

3. Why it is safe here, when it was not before

The obstruction in the earlier rows was never the ultraviolet. It was this: the $p^+ \rightarrow 0$ subtraction replaces $h(p^+)$ by $h(0)$, and if p^+ sits *inside* a transverse phase then setting $p^+ = 0$ **strips that oscillation**, leaving a bare $1/z^2$ tail that the phase had been damping. Here that cannot happen:

$$w - z = \frac{S}{k^+}(y - z) \implies w' - w = \frac{S}{k^+}(y' - y) \implies e^{-ik \cdot (w' - w)} = \underbrace{e^{-ik \cdot (y' - y)}}_{\text{no } p^+} \underbrace{e^{-ik p^+}}_{\text{no } z, x, x'}.$$

The two $\delta^{(2)}$'s moved every trace of z out of the phase. So:

- z, x, x' never see p^+ at all. Nothing about them changes under the subtraction.
- y, y' see p^+ only through $e^{-ik p^+}$; at $p^+ = 0$ the phase becomes $e^{-ik \cdot (y' - y)}$, **not 1**. The leading-order oscillation survives, so the Bessel $R^{-1/2}$ damping that makes the large- $|y|$ region converge is still there. Measured on the subtraction term itself:

	R	10^2	10^3	10^4
$ A(R) R^2$ at $p^+ = 0$		2.41×10^{-2}	1.04×10^{-2}	8.79×10^{-3}

Falling, consistent with the $R^{-1/2}$ Bessel envelope (the sampled values oscillate inside it). The subtraction term is transverse-convergent wherever the unsubtracted one is.

4. What leaving z un-integrated costs you

The $\frac{2}{k^+}(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km})$ term cancels only *after* $\int d^2z$, because that is when the tensor $T^{mm'} = \int d^2z \frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2}$ becomes symmetric. Pointwise in z it is

$$\frac{2}{k^+} \left[(M \cdot X') (N \cdot X) - (M \cdot X) (N \cdot X') \right], \quad M = \frac{y-z}{(y-z)^2}, \quad N = \frac{y'-z}{(y'-z)^2},$$

which is **not** zero: measured maximum 173.7 on the grid, integrating to 1.5×10^{-3} (i.e. zero, at quadrature accuracy). So carry it. It is harmless, but do not drop it or make any approximation to the z dependence before integrating, or you will keep a spurious remainder.

5. What the $+$ prescription does not do for you

It does not regulate z . The trace of the z integral is still $\pi \log \frac{R^2}{r^2}$ — at $R = 300$ in the test above, 33.24 — and that logarithm diverges as $R \rightarrow \infty$ at every value of p^+ , including $p^+ = 0$. You still need $d = 2 - 2\epsilon$ (or a cutoff) on the z integral.

The good news is that §2 says the two regulators commute, so you are free to:

1. apply $[\frac{1}{p^+}]_+$ now, keeping all five transverse integrals intact,
2. extract $\log \frac{\sqrt{-k^+}}{\Lambda}$ and hand it to JIMWLK — it multiplies the $p^+ \rightarrow 0$ limit of the transverse structure, which is the LO kernel times the real BK kernel,
3. regulate and do the z integral whenever it suits you.

The answer is the same either way.

5.1 Summary

question	answer
Can the $+$ prescription be applied with no transverse integration?	Yes
Does it introduce a new transverse divergence?	No — z has no phase at any p^+
Do p^+ and the transverse integrals commute?	Yes, verified numerically
Is the $p^+ \rightarrow 0$ subtraction transverse-convergent?	Yes — the LO phase survives
Can you then drop the $\frac{2}{k^+}$ term?	No — only zero after $\int d^2z$
Does it spare you the ϵ on z ?	No — the IR log is p^+ -independent

zint/src/row8_order.py, row8_order_conv.py — every number above.